Airline Passengers Satisfaction Predictions Using Classification Models

Module 8: Machine Learning – Classification

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Problem Statement

This project analyzes an airline passenger satisfaction dataset to:

- Identify the main factors influencing customer satisfaction.
- Explore patterns and insights through EDA.
- Build and evaluate classification models that predict if a passenger is either satisfied or is neutral/unsatisfied.
- Provide recommendations for improving customer experience.
- This analysis is valuable for airline customer experience teams, data analysts, and business decision-makers who want data-driven improvements in service quality.

Data Dictionary

- The link to an external data dictionary is Airline Satisfaction Data Dictionary.

- The original Dataset has 103,904 records and 25 variables.

- The clean Dataset (to apply the Classification Models) has 97,799 records and 25 variables.

Variable	Description
Gender	Gender of the passengers (Female, Male)
Customer Type	The customer type (Loyal customer, disloyal customer)
Age	The actual age of the passengers
Type of Travel	Purpose of the flight of the passengers (Personal Travel, Business Travel)
Class	Travel class in the plane of the passengers (Business, Eco, Eco Plus)
Flight distance	The flight distance of this journey
Inflight wifi service	Satisfaction level of the inflight wifi service (0:Not Applicable;1-5)
Departure/Arrival time convenient	Satisfaction level of Departure/Arrival time convenient
Ease of Online booking	Satisfaction level of online booking
Gate location	Satisfaction level of Gate location
Food and drink	Satisfaction level of Food and drink
Online boarding	Satisfaction level of online boarding
Seat comfort	Satisfaction level of Seat comfort
Inflight entertainment	Satisfaction level of inflight entertainment
On-board service	Satisfaction level of On-board service
Leg room service	Satisfaction level of Leg room service
Baggage handling	Satisfaction level of baggage handling
Check-in service	Satisfaction level of Check-in service
Inflight service	Satisfaction level of inflight service
Cleanliness	Satisfaction level of Cleanliness
Departure Delay in Minutes	Minutes delayed when departure
Arrival Delay in Minutes	Minutes delayed when Arrival
Satisfaction	Airline satisfaction level(Satisfaction, neutral or dissatisfaction)

Executive Summary

After cleaning and preparing the data, the final dataset contains 97,799 records and 23 variables (22 Features and 1 Target - Satisfaction).

Key Findings:

- There are no missing values in the dataset, except for Arrival Delay in minutes.
- There are no duplicate values.
- The categorical values need to be converted to numeric as described in the data cleaning steps.
- The outliers are possible values within the industry.

Data Cleaning Steps

- Removed missing values in the arrival delay column.

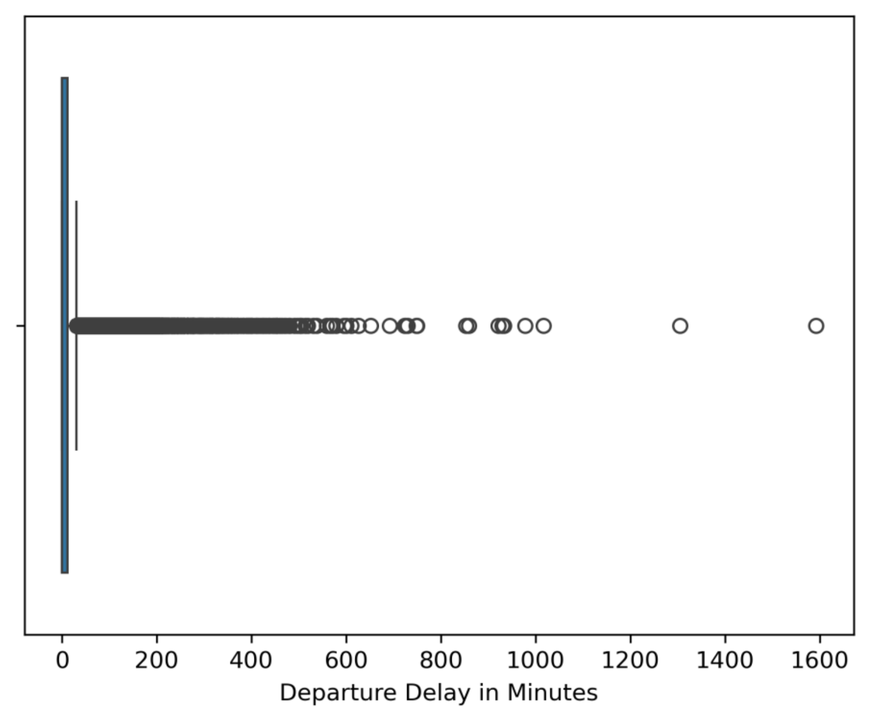
- Converted categorical variables using one-hot encoding for Gender, Customer Type, Type of Travel, Class and satisfaction.

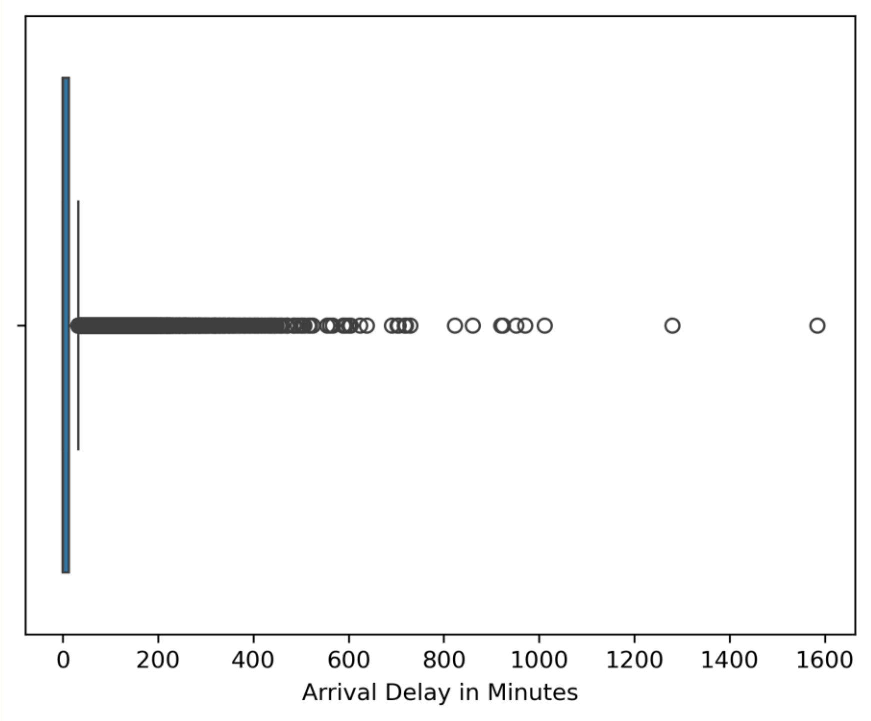
- Checked for anomalies and outliers in delay-related fields:

    - Most flights have very small delays (median = 0).
    - Few have huge delays (hundreds or >1000 minutes).
    - Outliers are real operational events, such diversions or technical failures, not errors.
    - However, because KNN uses distances, extreme values will “pull” the distance space and distort neighbors.
    - The Outliers removal will be removing the extreme values up to the 95th Perc.
    - Class balance 56% for neutral/unsatisfied and 44% for satisfied passengers.

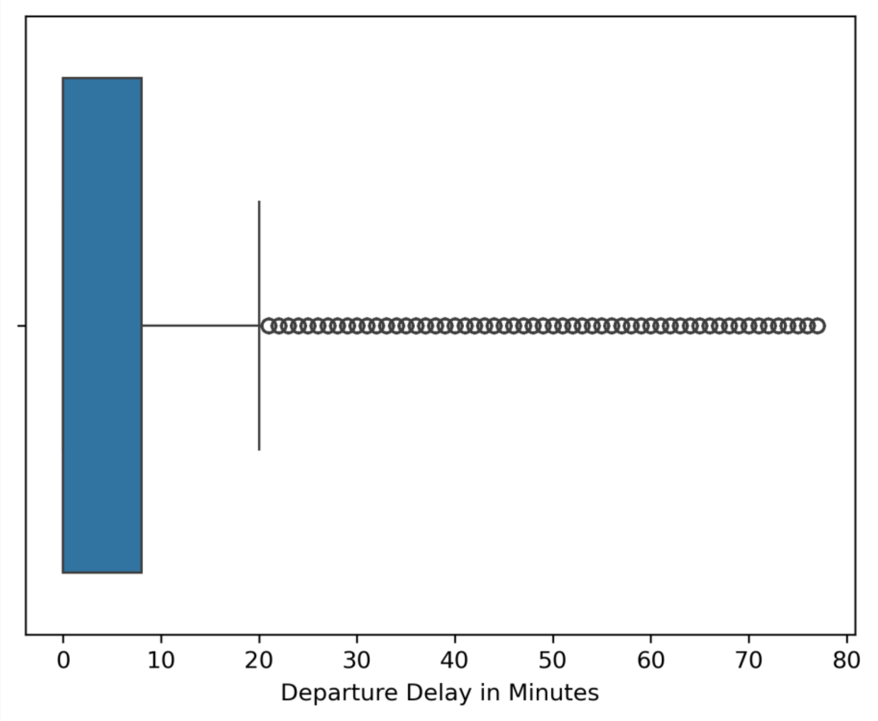
- Scaled continuous variables for KNN and Logistic Regression Models.

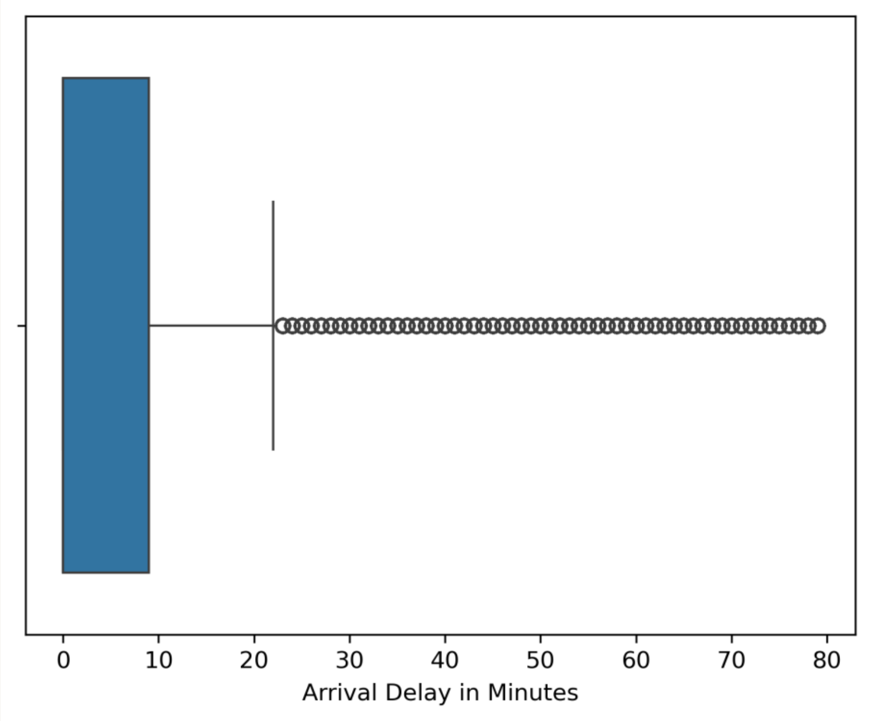
EDA – Visualization 1 – Boxplots of original dataset Departure and Arrival Delays





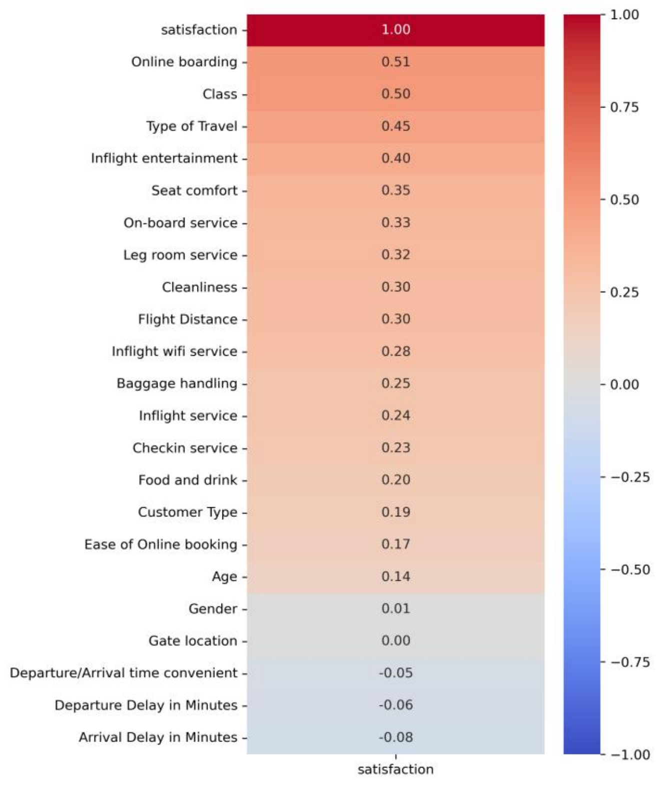
EDA – Visualization 2 – Boxplots after outliers removal of Departure and Arrival Delays





EDA – Visualization 3 – Correlation Heatmap w/Target

- Shows the feature correlation with Satisfaction.
- The Features Gender and Gate location have the weakest correlation to satisfaction.
- The strongest correlations features are Online boarding, Class, Type of Travel, Inflight entertainment and Seat comfort and are also shown in the Pair Plot



EDA – Visualization 4 – Pairplots of Highly Correlated Vars

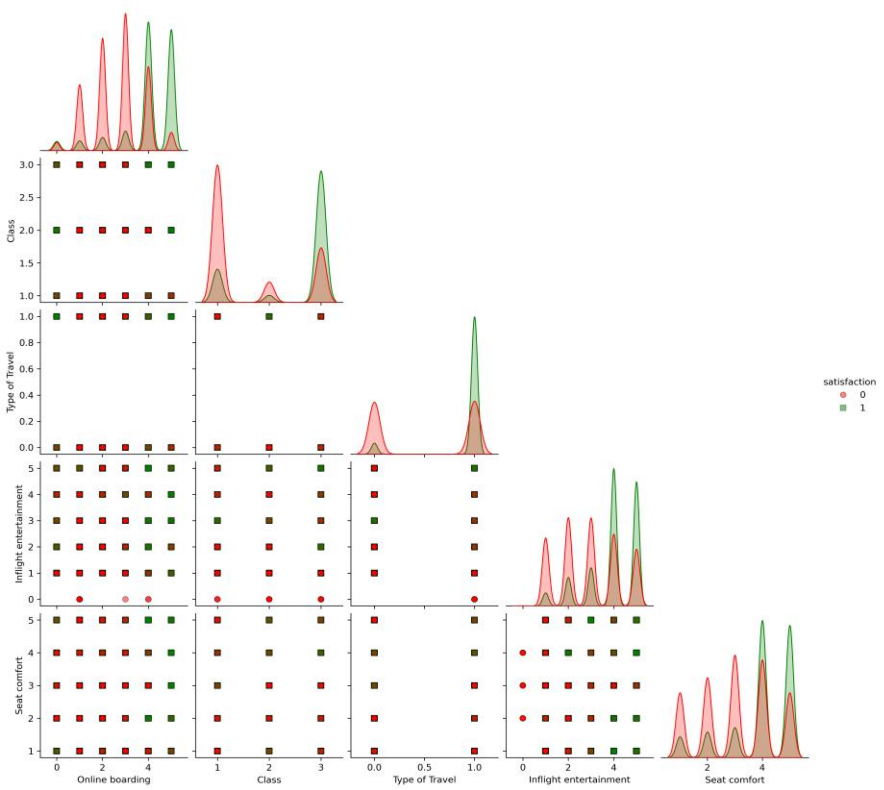
- The scatterplots look discrete because the values of the Features are or Ordinal.

- For KNN, the distance metric will still work fine — the discreteness is just a visualization.

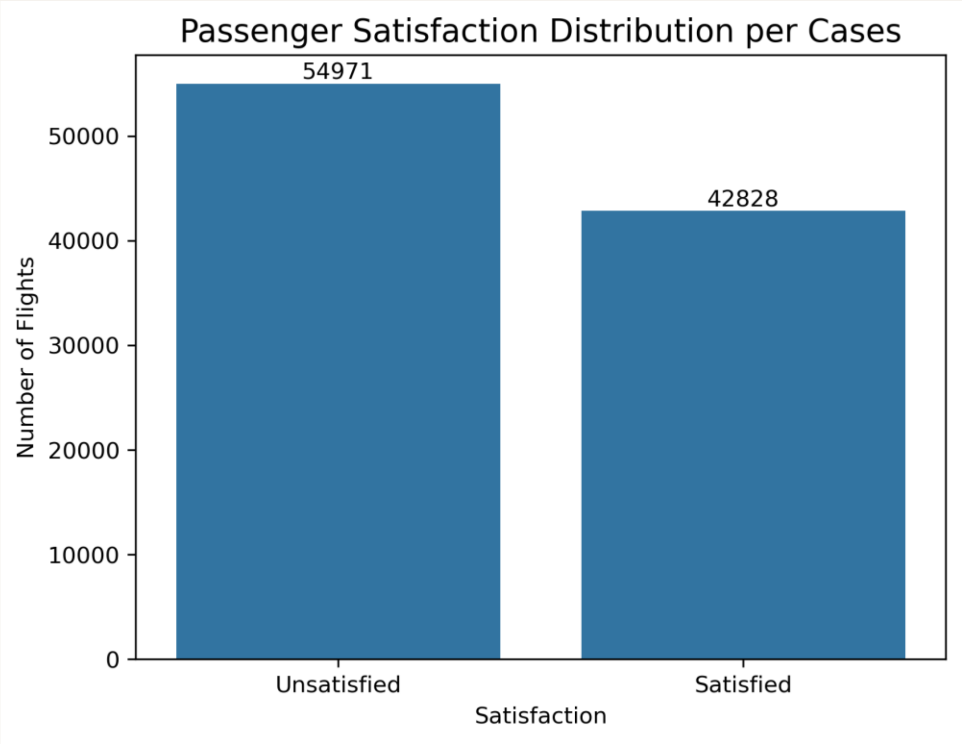
- There are areas where the dots are “redder”, indicating dissatisfied customers, or "greener", indicating satisfied customers.

- In general, higher features such Class, Flight Entertainment and Seat Comfort have a higher level of satisfaction.

- Business Travel has also a higher satisfaction level.



EDA – Visualization 5 – Class Balance



- **56% for Neutral or Unsatisfied Passengers**
- **44% for Satisfied Passengers**

Model Performance

Model Selection: The project tested several classification algorithms to identify the best model for predicting passenger satisfaction:

- Baseline Accuracy Score of 0.5621.
- KNN with default hyperparameters.
- KNN with optimized hyperparameters.
- Logistic Regression – baseline interpretable model.
- Random Forest Classifier – robust for tabular data, handles nonlinearity

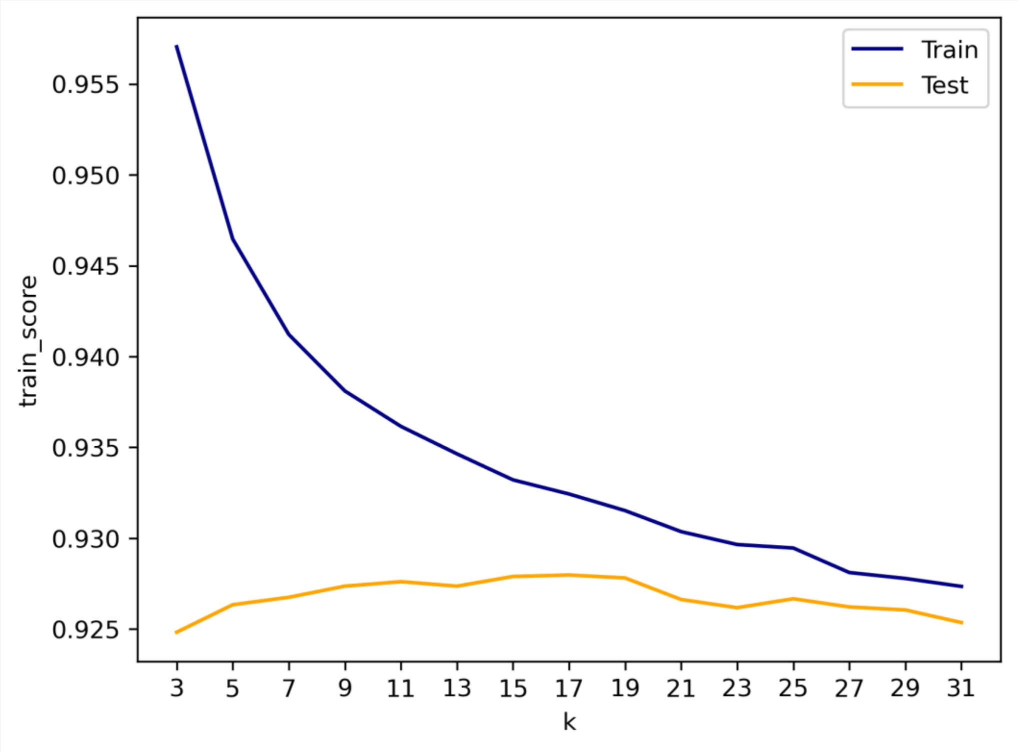
Model Performance

Evaluation Metrics:

Model	Accuracy	False Positives	False Negatives	Quick interpretation
KNN (k=5)	0.9263	525	1276	Great Accuracy, more FNs than FPs, okay for the business model
KNN (k=17)	0.9280	430	1331	Slightly better accuracy than KNN (default), lower FPs than default k
Logistic Regression	0.8743	1292	1693	The worst model in both Accuracy and Errorss
Random Forest	0.9625	299	618	The best model in both Accuracy and Errors

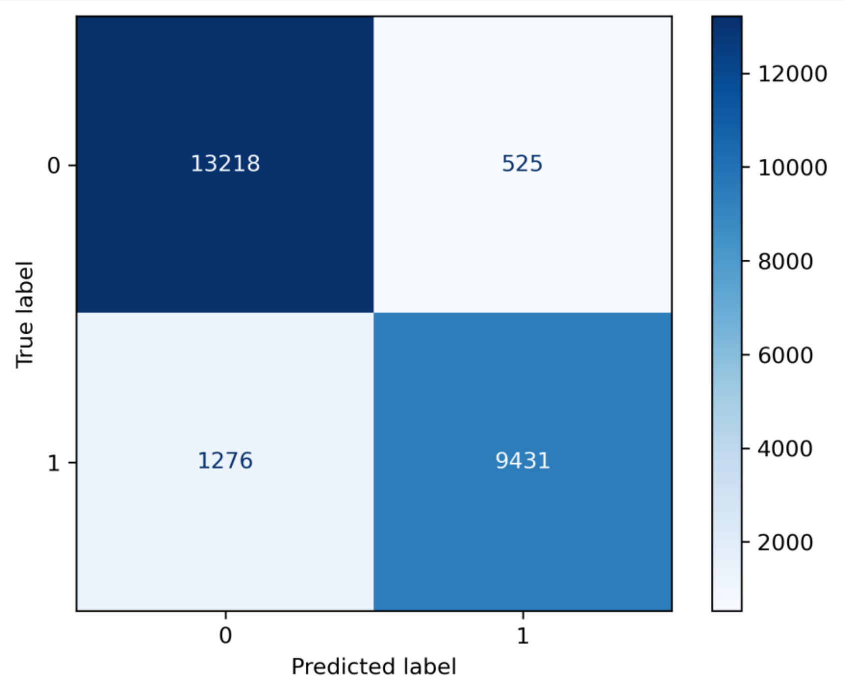
All models are valid because they beat the Baseline Accuracy Score. Random Forest is the best model, having the highest accuracy and the lowest errors.

KNN Model Hyperparameter Optimization

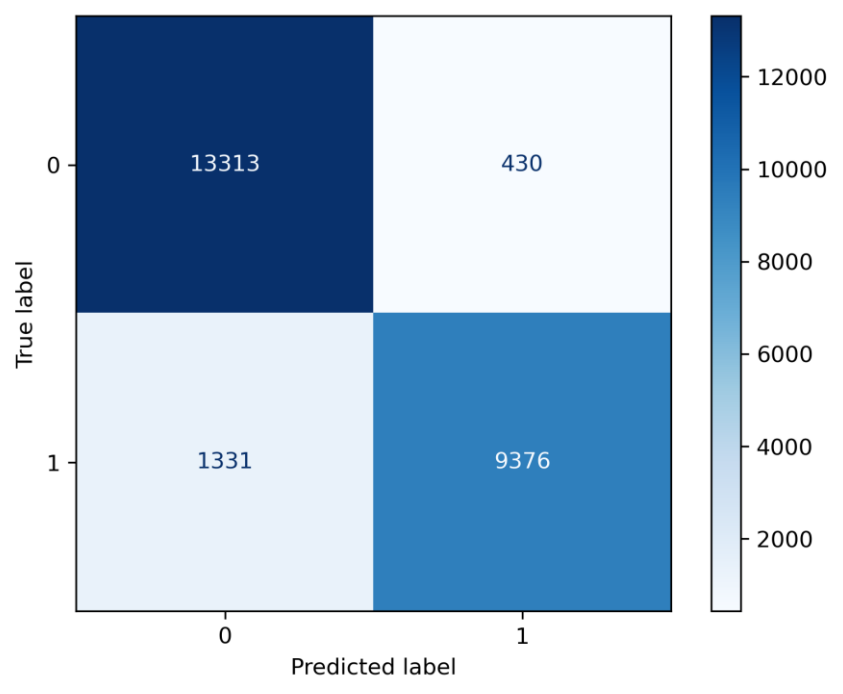


K=17 (optimized)

KNN Model Confusion Matrices

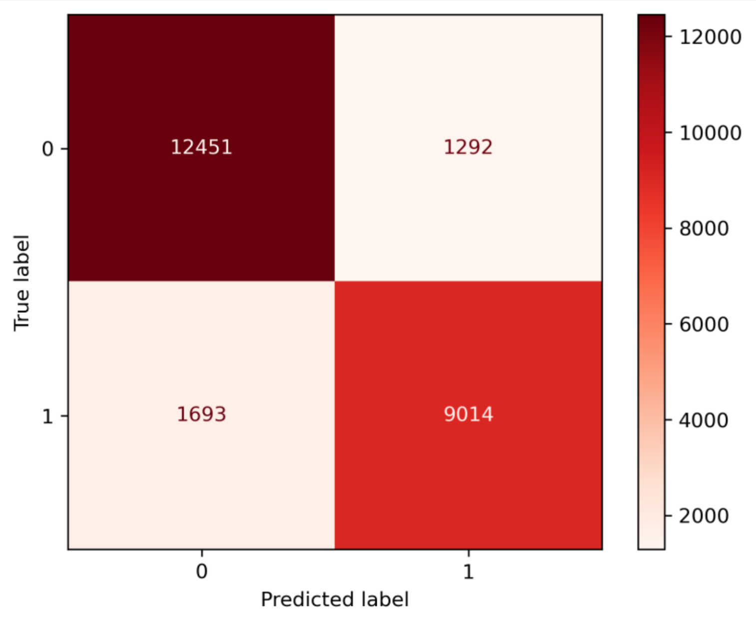


K=5 (default)

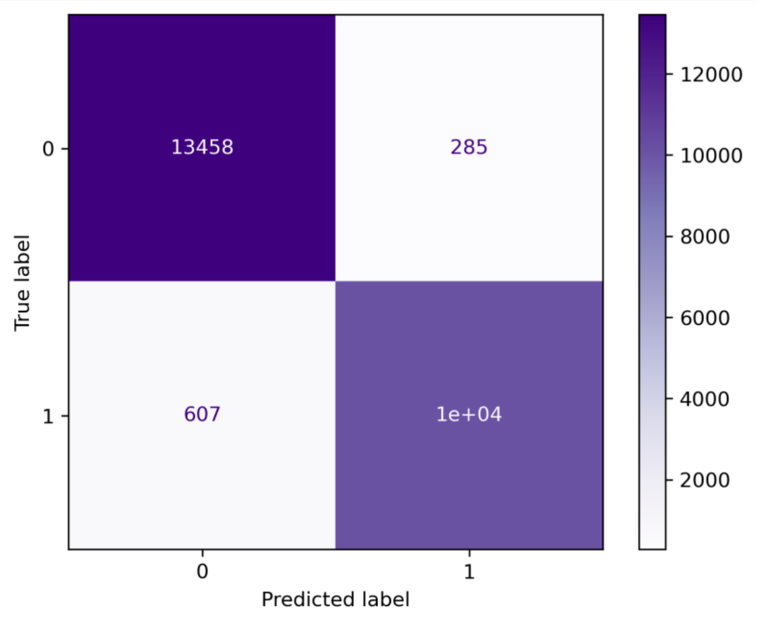


K=17 (optimized)

Logistic Regression and Random Forest Confusion Matrices



Logistic Regression

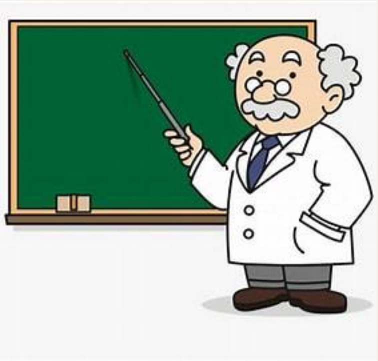


Random Forest

Conclusions

- All models have good Accuracy > 0.87, much better than the 0.52 of the Baseline Model.
- The best model by FN, FP and Accuracy is Random Forest.
- Both KNN models have better scores and accuracy than Logistic Regression.
- Adjusting the k-value on KNN models can increase Accuracy and decreased FPs, desirable for business purposes.
- The analysis explicitly states the model's strong performance relative to baseline.
- It ties the FN vs FP pattern to business impact, not arbitrary preference.
- It also frames FN as "unnecessary but harmless," and FP as "risky," which is usually accurate for customer-experience applications and justifies keeping the model despite the imbalance between FNs and FPs.

Next Steps and Recommendations



- Analyze the cost of handling the highest correlated features with satisfaction, such online boarding, inflight entertainment and seat comfort, especially for Eco class.
- Improve boarding process efficiency.
- Invest in seat comfort and inflight entertainment upgrades.
- Enhance consistency in on-board service interactions.
- Improve as much as possible the above features for Eco and Eco Plus classes.
- Compare the cost of these measures to the potential revenue loss of unsatisfied passengers.

Thank You

- Any Questions?
- Please feel free to reach through Disco Messaging
- Thanks for attending, take care!!